EXAM TOOLS

On Tips Notes

Great achievements start with the smallest steps—and the right guidance can make all the difference. At Oswaal, we believe that every learner has the potential to excel when concepts are crystal clear and learning is simplified. That's why we've created our ON TIPS NOTES—your quick, powerful tool to master every chapter from the syllabus.

These notes are designed to spark confidence, strengthen understanding and ensure that no concept is left behind. Keep them close, revise smart and walk into every exam ready to win. With Oswaal by your side, success isn't just possible—it's inevitable.

1. RELATIONS AND FUNCTIONS

FUNDAMENTALS

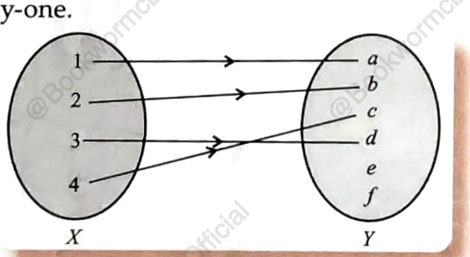
- A relation R from a non-empty set A to a non-empty set B is a subset of the Cartesian product $A \times B$.
- The set of all the first elements of the ordered pairs in a relation R from a set A to a set B is called the domain of the relation R.
- The set of all second elements in a relation R from a set A to a set B is called the range of the relation R.
- The whole set B is called the co-domain of the relation R.

TYPES OF RELATION

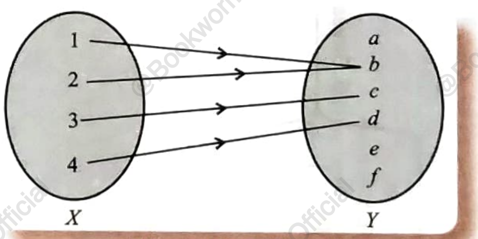
- **Empty Relation :** A relation R in a set A is called empty relation, if no element of A is related to any element of A, i.e., $R = \phi \subset A \times A$.
- **Universal Relation :** A relation R in a set A is called universal relation, if each element of A is related to every element of A, i.e., $R = A \times A$.
- A relation R in a set A is called
  (i) reflexive, if $(a, a) \in R$, for every $a \in A$.
  (ii) symmetric, if $(a_1, a_2) \in R \Rightarrow (a_2, a_1) \in R, \forall\, a_1, a_2 \in A$.
  (iii) transitive, if $(a_1, a_2) \in R$ and $(a_2, a_3) \in R \Rightarrow (a_1, a_3) \in R, \forall\, a_1, a_2, a_3 \in A$.
- **Equivalence Relation :** A relation R in a set A is said to be an equivalence relation if R is reflexive, symmetric and transitive.

TYPES OF FUNCTION

- **One-One (Injective) and Many-One :** A function $f : X \rightarrow Y$ is defined to be one-one (or injective), if the images of distinct elements of X have distinct images in Y under f, i.e., $\forall\, x_1, x_2 \in X, f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. Otherwise, f is called many-one.

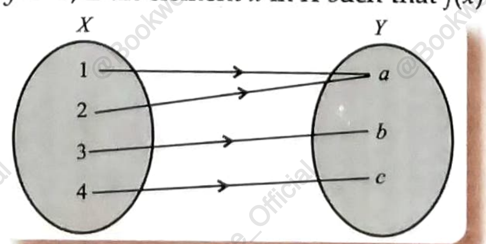


One-One Function



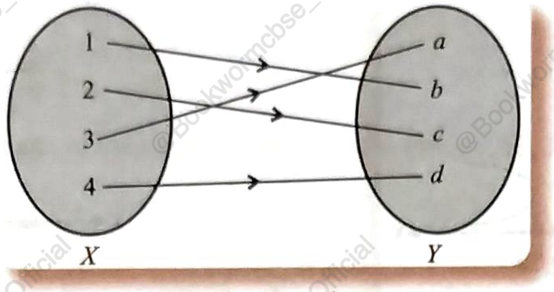
Many-One Function

- **Onto (Surjective) :** A function $f : X \rightarrow Y$ is said to be onto (or surjective), if every element of Y is the image of some element of X under f, i.e., $\forall\, y \in Y, \exists$ an element x in X such that $f(x) = y$.



Onto Function

- **One-One and Onto (Bijective) :** A function $f : X \to Y$ is said to be one-one and onto (or bijective), if f is both one-one and onto.



Bijective Function

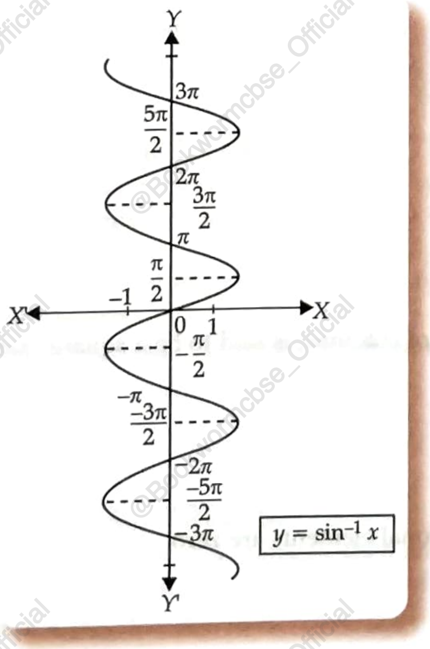
2. INVERSE TRIGONOMETRIC FUNCTIONS

THE PRINCIPAL VALUE BRANCH OF TRIGONOMETRIC INVERSE FUNCTIONS IS AS FOLLOWS:

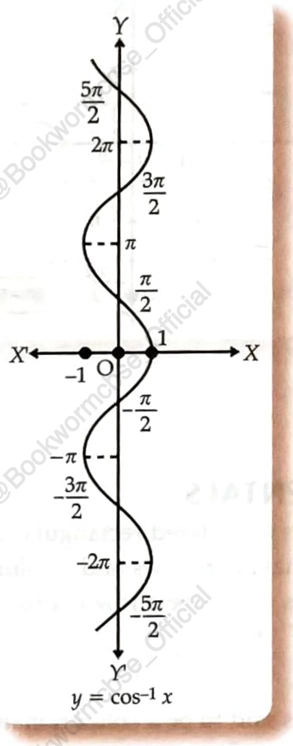
Inverse Function	Domain	Principal Value Branch
$\sin^{-1}$	$[-1, 1]$	$\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1}$	$[-1, 1]$	$[0, \pi]$
cosec^{-1}	$\mathbb{R} - (-1, 1)$	$\left[\frac{-\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
$\sec^{-1}$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$
$\tan^{-1}$	$\mathbb{R}$	$\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$
$\cot^{-1}$	$\mathbb{R}$	$(0, \pi)$

GRAPHS OF INVERSE TRIGONOMETRIC FUNCTIONS

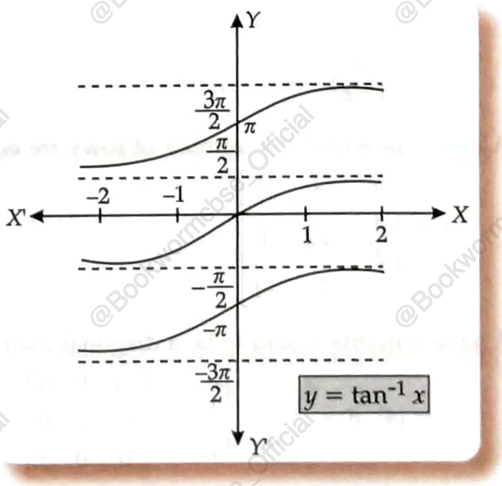
1. Graph of $\sin^{-1} x$



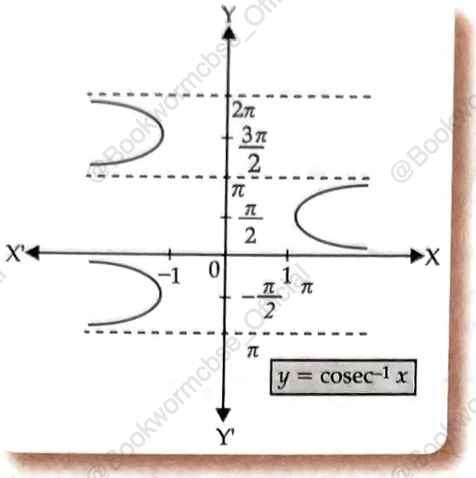
2. Graph of $\cos^{-1} x$



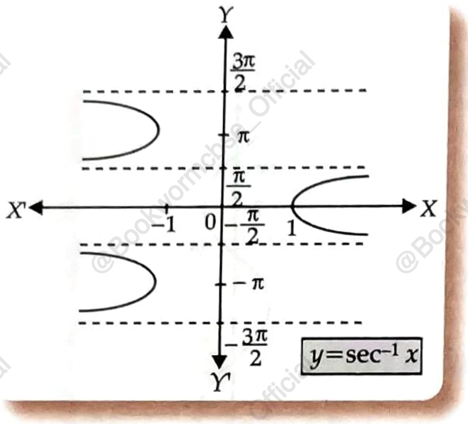
3. Graph of $\tan^{-1} x$



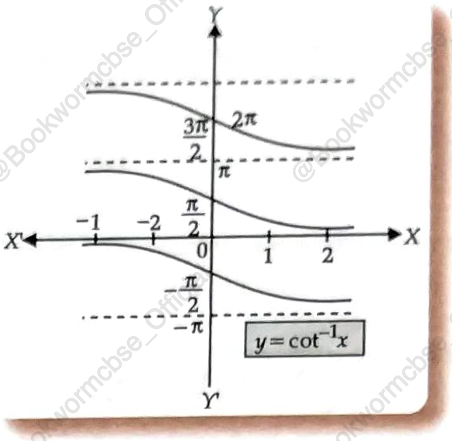
4. Graph of $\text{cosec}^{-1} x$



5. Graph of $\sec^{-1} x$



6. Graph of $\cot^{-1} x$



3. MATRICES

FUNDAMENTALS

- A matrix is an ordered rectangular array of numbers (or functions).
- A matrix having m rows and n columns is called a matrix of order $m \times n$.
- A matrix is said to be a row matrix if it has only one row.

$$B = \begin{bmatrix} -\frac{1}{2} & \sqrt{5} & 2 & 3 \end{bmatrix}$$

- A matrix is said to be a column matrix if it has only one column.

$$A = \begin{bmatrix} 0 \\ \sqrt{3} \\ -1 \\ 1/2 \end{bmatrix}$$

- A matrix in which the number of rows are equal to the number of columns, is said to be a square matrix.

$$A = \begin{bmatrix} 3 & -1 & 0 \\ \frac{3}{2} & 3\sqrt{2} & 1 \\ 4 & 3 & -1 \end{bmatrix}$$

- A square matrix is said to be a diagonal matrix if its all non-diagonal elements are zero.

$$A = [4],\ B = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix},\ C = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

- A diagonal matrix is said to be a scalar matrix if its diagonal elements are equal.

$$A=[3],\ B=\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix},\ C=\begin{bmatrix}\sqrt{3} & 0 & 0\\ 0 & \sqrt{3} & 0\\ 0 & 0 & \sqrt{3}\end{bmatrix}$$

- A square matrix in which elements in the diagonal are all 1 and rest are all zeroes is called an identity matrix. Identity matrix is denoted by I.

$$A=[1],\ B=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix},\ C=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

- A matrix is said to be a zero matrix or null matrix if all its elements are zeroes.

$$A=[0],\ B=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix},\ C=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix},\ D=[0\quad 0]$$

- A square matrix A = $[a_{ij}]_{m\times m}$ is said to be an identity matrix

$$\text{if } a_{ij}=\begin{cases}1, & \text{if } i=j\\ 0, & \text{if } i\neq j\end{cases}$$

$$I_3=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}_{3\times 3},\ I_2=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}_{2\times 2}$$

- Two matrices can be added if they are of the same order.
- If A, B are respectively $m\times n$, $k\times l$ matrices, then both AB and BA are defined if and only if $n=k$ and $l=m$.

TRANSPOSE OF A MATRIX

- If $A=[a_{ij}]$ be an $m\times n$ matrix, then the matrix obtained by interchanging the rows and columns of A is called the transpose of A.
- If $A=\begin{bmatrix}3 & 5\\ \sqrt{3} & 1\\ 0 & \frac{-1}{5}\end{bmatrix}_{3\times 2}$, then $A'=\begin{bmatrix}3 & \sqrt{3} & 0\\ 5 & 1 & \frac{-1}{5}\end{bmatrix}_{2\times 3}$
- Properties:
  (i) $(A^T)^T=A$
  (ii) $(kA)^T=kA^T$ (where k is any constant)
  (iii) $(A+B)^T=A^T+B^T$
  (iv) $(AB)^T=B^TA^T$

SYMMETRIC AND SKEW SYMMETRIC MATRICES

(i) A square matrix $A=[a_{ij}]$ is said to be symmetric if $A^T=A$, i.e., $a_{ij}=a_{ji}$ for all possible values of i and j.
(ii) A square matrix $A=[a_{ij}]$ is said to be a skew symmetric matrix if $A^T=-A$, i.e. $a_{ij}=-a_{ji}$ for all possible values of i and j.

- **Theorem 1 :** For any square matrix A with real number entries, $A+A^T$ is a symmetric matrix and $A-A^T$ is a skew symmetric matrix.
- **Theorem 2 :** Any square matrix A can be expressed as the sum of a symmetric matrix and a skew symmetric matrix, i.e.,

$$A=\frac{(A+A^T)}{2}+\frac{(A-A^T)}{2}$$

EQUALITY OF MATRICES:

- If A and B are two $m\times n$ matrices, then another $m\times n$ matrix obtained by adding the corresponding elements of the matrices A and B is called the sum of the matrices A and B and is denoted by '$A+B$'.
- Thus if $A=[a_{ij}]$, $B=[b_{ij}]$, or $A+B=[a_{ij}+b_{ij}]$

MULTIPLICATION OF A MATRIX BY A SCALAR:

If a $m \times n$ matrix A is multiplied by a scalar k (say), then the new kA matrix is obtained by multiplying each element of matrix A by scalar k. Thus, if $A = [a_{ij}]$ and it is multiplied by a scalar k, then $kA = [ka_{ij}]$, i.e. $A = [a_{ij}]$ or $kA = [ka_{ij}]$.

e.g., $$A = \begin{bmatrix} 2 & -4 \\ 5 & 6 \end{bmatrix} \text{ or } 3A = \begin{bmatrix} 6 & -12 \\ 15 & 18 \end{bmatrix}$$

MULTIPLICATION OF TWO MATRICES:

Let $A = [a_{ij}]$ be a $m \times n$ matrix and $B = [b_{jk}]$ be a $n \times p$ matrix such that the number of columns in A is equal to the number of rows in B, then the $m \times p$ matrix $C = [c_{ik}]$ such that $[c_{ik}] = \sum_{j=1}^{n} a_{ij}\, b_{jk}$ is said to be the product of the matrices A and B in that order and it is denoted by AB i.e., "$C = AB$".

4. DETERMINANTS

FUNDAMENTALS

- Only square matrices have determinants.
- For a matrix A, $|A|$ is read as determinant of A and not as modulus of A.

DETERMINANT OF A MATRIX OF ORDER ONE

Let $A = [a]$ be the matrix of order 1, then determinant of A is defined to be equal to a.

DETERMINANT OF A MATRIX OF ORDER TWO

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$

DETERMINANT OF A MATRIX OF ORDER THREE

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}(a_{22}\,a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{31}\,a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22})$$

MINORS

Minor of an element a_{ij} of a determinant (*or* a determinant corresponding to matrix A) is the determinant obtained by deleting its i^{th} row and j^{th} column in which a_{ij} lies. Minor of a_{ij} is denoted by M_{ij}. Hence, we can get 9 minors corresponding to the 9 elements of a third order (i.e., 3×3) determinant.

CO-FACTORS

Cofactor of an element a_{ij}, denoted by A_{ij}, is defined by $A_{ij} = (-1)^{(i+j)} M_{ij}$, where M_{ij}, is minor of a_{ij}. Sometimes C_{ij} is used in place of A_{ij} to denote the co-factor of element a_{ij}.

AREA OF A TRIANGLE

Area of a triangle with vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3) is given by

$$\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

INVERSE OF A MATRIX

Step 1 : Find out Minor for every element.

Minor of an element a_{ij} of the determinant of matrix A is the determinant obtained by deleting i^{th} row and j^{th} column and it is denoted by M_{ij}.

e.g., Minor of a_{21} of $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

$$M_{21} = \begin{vmatrix} - & a_{12} & a_{13} \\ - & - & - \\ - & a_{32} & a_{33} \end{vmatrix}$$

$$= -\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

Step 2 : Find out Co-factors of every Minor.

Co-factor of an element a_{ij} is given by $A_{ij} = (-1)^{i+j} M_{ij}$.

Step 3 : Find out Adjoint.

The adjoint of a square matrix is the transpose of matrix of Co-factors.

Step 4 : Find out determinant of A i.e., $|A|$.

Step 5 : Find out inverse.

$$A^{-1} = \frac{1}{|A|} adj\ A$$

HOW TO SOLVE LINEAR SYSTEM OF EQUATIONS?

Consider the system of equations

$a_1x + b_1y + c_1z = d_1$

$a_2x + b_2y + c_2z = d_2$

$a_3x + b_3y + c_3z = d_3$

$$\text{Let } A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

The given system of equations can be written as :

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \text{ or } AX = B$$

$\therefore \quad X = A^{-1}B$

CONSISTENT SYSTEM:

A system is considered to be consistent if it has at least one solution.

INCONSISTENT SYSTEM:

If a system has no solution, it is said to be inconsistent.

5. CONTINUITY AND DIFFERENTIABILITY

CONTINUITY (DEFINITION):

- If f is a real valued function on a subset of real numbers and let c be a point in its domain, then f is a continuous function at c i.e.,

$$\lim_{x\to c^+} f(x) = f(c) = \lim_{x\to c^-} f(x)$$

- Obviously, if left hand limit and right hand limit and value of the function at $x = c$ exists and are equal to each other then f is continuous at $x = c$.

ALGEBRA OF CONTINUOUS FUNCTIONS:

- Let f and g be two real functions continuous at $x = c$, then
  - **(i)** Sum of two functions is continuous at $x = c$ i.e., $(f + g)(x)$ is defined as $f(x) + g(x)$ is continuous at $x = c$.
  - **(ii)** Difference of two functions is continuous at $x = c$ i.e., $(f - g)(x)$ is defined as $f(x) - g(x)$ is continuous at $x = c$.
  - **(iii)** Product of two functions is continuous at $x = c$ i.e., $(fg)(x)$ is defined as $f(x).g(x)$ is continuous at $x = c$.
  - **(iv)** Quotient of two functions is continuous at $x = c$. (Provided it is defined at $x = c$)

    i.e., $\left(\frac{f}{g}\right)(x)$ is defined as $\frac{f(x)}{g(x)}[g(x) \neq 0]$ is continuous at $x = c$.

However, if $f(x) = \lambda$, then

(a) $\lambda.g$ is defined by $\lambda.g(x)$, then $\lambda.g$ is also continuous at $x = c$.

(b) Similarly, if $\frac{\lambda}{g}$ is defined as $\frac{\lambda}{g}(x) = \frac{\lambda(x)}{g(x)}$, then $\frac{\lambda}{g}$ is also continuous at $x = c$. $[g(x) \neq 0]$

DIFFERENTIABILITY (DEFINITION):

Let f be a real function and c is a point in its domain. The derivative of f at c is defined as

$\lim_{h\to 0} \frac{f(c+h)-f(c)}{h}$, provided limit exists

$$= f'(c) = \left[\frac{d}{dx}f(x)\right]_{x=c}$$

$f'(x)$ is defined as $$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

wherever the limit exists, is defined to be the derivative of f. It is denoted as :

$$f'(x) \text{ or } \frac{d}{dx}[f(x)]$$

or if $f(x) = y$, then $$f'(x) = \frac{dy}{dx}.$$

Every differentiable function is continuous.

ALGEBRA OF DERIVATIVES:

Let u, v be the functions of x.

(i) $(u \pm v)' = u' \pm v'$

(ii) $(uv)' = u'v + uv'$ (Leibnitz or Product Rule)

(iii) $\left(\frac{u}{v}\right)' = \frac{u'v-uv'}{v^2}$, where $v \neq 0$. (Quotient Rule)

DERIVATIVES OF INVERSE TRIGONOMETRIC FUNCTIONS :

Function	Domain	Derivative
$\sin^{-1}x$	[–1, 1]	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1}x$	[–1, 1]	$-\frac{1}{\sqrt{1-x^2}}$
$\tan^{-1}x$	$\mathbb{R}$	$\frac{1}{1+x^2}$
$\cot^{-1}x$	$\mathbb{R}$	$-\frac{1}{1+x^2}$
$\sec^{-1}x$	$\mathbb{R}-(-1, 1)$	$\frac{1}{x\sqrt{x^2-1}}$
$\text{cosec}^{-1}x$	$\mathbb{R}-(-1, 1)$	$-\frac{1}{x\sqrt{x^2-1}}$

IMPLICIT FUNCTIONS :

An equation in the form of $f(x, y) = 0$ in which y is not expressible in terms of x is called as an implicit function of x and y. Both sides of the equation are differentiated term wise, then from this equation, $\frac{dy}{dx}$ is obtained. It may be noted that when a function of y occurs, then differentiate it w.r.t. y and multiply it by $\frac{dy}{dx}$. e.g., To find $\frac{dy}{dx}$ from $\cos^2 y + \sin xy = 1$, we differentiate it as:

$$2\cos y(-\sin y)\frac{dy}{dx}+\cos(xy).\frac{d}{dx}(xy) = 0$$

or $$-2\sin y\cos y\frac{dy}{dx}+\cos xy\left(x\frac{dy}{dx}+y\right) = 0$$

or $$\frac{dy}{dx} = \frac{y\cos xy}{2\sin y\cos y-x\cos(xy)}$$

EXPONENTIAL FUNCTIONS:

The exponential function with positive base $b > 1$, is the function $y = b^x$.

(i) The graph of $y = 10^x$ is shown in figure.

(ii) Domain $= \mathbb{R}$.

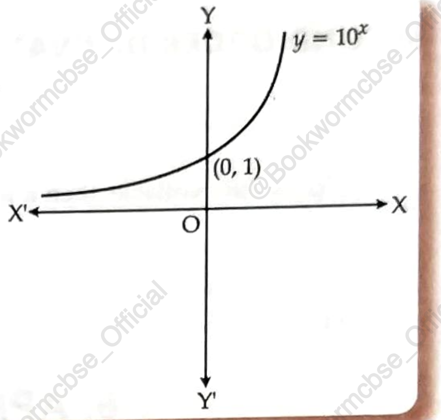
(iii) Range $= \mathbb{R}^+$. (all the real numbers)

(iv) The point (0, 1) always lies on the graph.

(v) It is an increasing function i.e., as we move from left to right, the graph rises above.

(vi) As $x \to -\infty$, $y \to 0$.

(vii) $\frac{d}{dx}(a^x)=a^x\log_e a,\ \frac{d}{dx}(e^x)=e^x.$



LOGARITHMIC FUNCTIONS :

Let $b > 1$ be real number, then $b^x = a$ may be written as $\log_b a = x$.

(i) The graph of $y = \log_{10} x$ is shown in the figure.

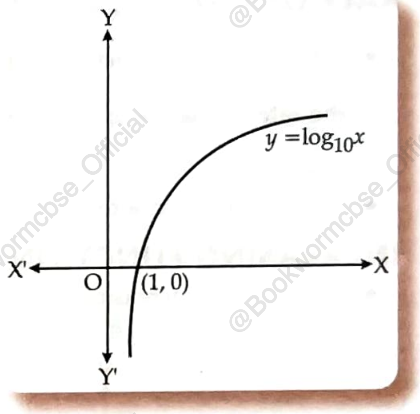
(ii) Domain $= \mathbb{R}^+$, Range $= \mathbb{R}$.

(iii) It is an increasing function.

(iv) As $x \to 0$, $y \to -\infty$.

(v) The function $y = e^x$ and $y = \log_e x$ are the mirror images of each other in the line $y = x$.

(vi) $\frac{d}{dx}(\log_a x)=\frac{1}{x}\log_a e, \frac{d}{dx}(\log_e x)=\frac{1}{x}.$



OTHER PROPERTIES OF LOGARITHM:

(i) $\log_b pq = \log_b p + \log_b q$

(ii) $\log_b \frac{p}{q}=\log_b p-\log_b q$

(iii) $\log_b p^x = x \log_b p$

(iv) $\log_a b=\frac{\log_b b}{\log_b a}=\frac{1}{\log_b a}$

LOGARITHMIC DIFFERENTIATION:

Whenever the functions are given in the form $y = [u(x)]^{v(x)}$

Using chain rule

$$\log y = v(x)\log u(x)$$

$$\frac{dy}{dx} = y\left[\frac{v(x)}{u(x)}.u'(x)+v'(x)\cdot\log u(x)\right]$$

Example :

Let, $y = (\cos x)^{\sin x}$

Take log on both sides, simplify and differentiate

$$\log y = \log (\cos x)^{\sin x}$$

$$\log y = \sin x \log \cos x$$

$$\frac{1}{y}\frac{dy}{dx} = \cos x\log\cos x+\sin x\left(\frac{-\sin x}{\cos x}\right)$$

$\therefore$ $$\frac{dy}{dx} = (\cos x)^{\sin y}[\cos x\log\cos x-\sin x\tan x]$$

DERIVATIVES OF FUNCTIONS OF PARAMETRIC FORM :

Let the given equations be $x = f(t), y = g(t)$,
where t is called the parameter.

$$\frac{dx}{dt} = f'(t), \frac{dy}{dt} = g'(t)$$

$$\therefore \quad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{g'(t)}{f'(t)}$$

SECOND ORDER DERIVATIVE :

Let, $y = f(x)$, then

$$\frac{dy}{dx} = f'(x)$$

If $f'(x)$ is differentiable, then it is again differentiated.

$$\text{L.H.S.} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2} = D^2y = y'' = y_2$$

and

$$\text{R.H.S.} = \frac{d}{dx}[f'(x)] = f''(x).$$

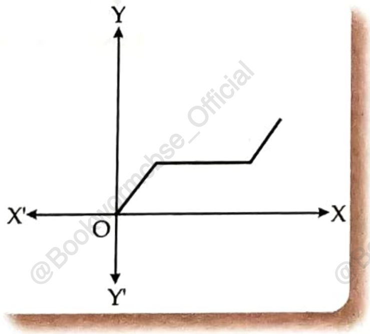
6. APPLICATIONS OF DERIVATIVES

RATE OF CHANGE OF QUANTITIES :

- Let $y = f(x)$ be a function. If the change in one quantity y varies with another quantity x, then $\frac{dy}{dx} = f'(x)$ denotes the rate of change of y with respect to x. At $x = x_0$, $\left.\frac{dy}{dx}\right|_{x=x_0}$ of $f'(x)$ represents the rate of change of y w.r.t. x at $x = x_0$.
- Let I be the open interval contained in the domain of real value function f.

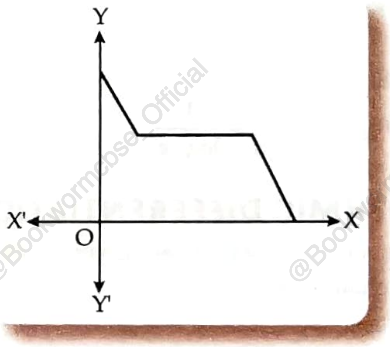
INCREASING FUNCTION :

- f is said to be increasing on I if $x_1 < x_2$ on I, then $f(x_1) \leq f(x_2)$ for all $x_1, x_2 \in I$.



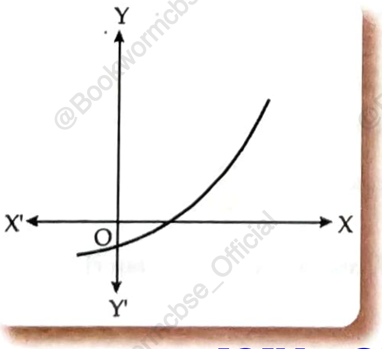
DECREASING FUNCTION :

- f is said to be decreasing function on I if $x_1 < x_2$ in I, then $f(x_1) \geq f(x_2)$ for all $x_1, x_2 \in I$.



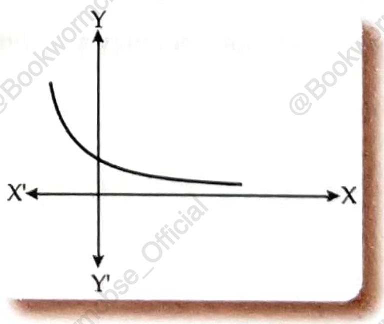
STRICTLY INCREASING FUNCTION :

- f is said to be strictly increasing on I if $x_1 < x_2$ in $I \Rightarrow f(x_1) < f(x_2)$ for all $x_1, x_2 \in I$.



STRICTLY DECREASING FUNCTION :

- f is said to be strictly decreasing function on I if $x_1 < x_2$ in $I \Rightarrow f(x_1) > f(x_2)$ for all $x_1, x_2 \in I$.



INCREASING AND DECREASING FUNCTION AT X_0 :

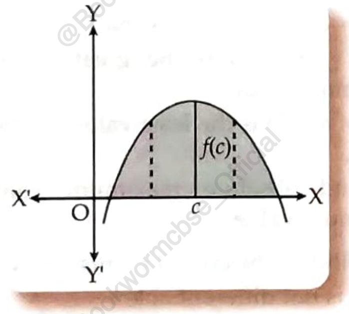
- Let x_0 be a point in the domain of definition of a real valued function f and there exists an open interval $I = (x_0 - h, x_0 + h)$ containing x_0 such that
  - **(i)** f is increasing at x_0, if $x_1 < x_2$ in $I \Rightarrow f(x_1) \le f(x_2)$.
  - **(ii)** f is strictly increasing at x_0, if $x_1 < x_2$ in $I \Rightarrow f(x_1) < f(x_2)$.
  - **(iii)** f is decreasing at x_0, if $x_1 < x_2$ in $I \Rightarrow f(x_1) \ge f(x_2)$.
  - **(iv)** f is strictly decreasing at x_0, if $x_1 < x_2$ $\Rightarrow f(x_1) > f(x_2)$.

TEST : INCREASING/DECREASING/ CONSTANT FUNCTION :

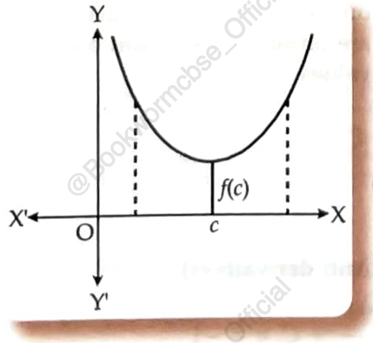
- Let f be a continuous function on $[a, b]$ and differentiable in an open interval (a, b), then
  - **(i)** f is increasing on $[a, b]$ if $f'(x) > 0$ for each $x \in (a, b)$.
  - **(ii)** f is decreasing on $[a, b]$ if $f'(x) < 0$ for each $x \in (a, b)$.
  - **(iii)** f is constant on $[a, b]$ if $f'(x) = 0$ for each $x \in (a, b)$.

MAXIMUM VALUE, MINIMUM VALUE, EXTREME VALUE :

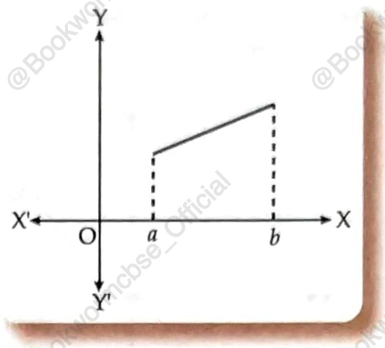
- Let f be a function defined in the interval I, then
  - **(i) Maximum value :** If there exists a point c in I such that $f(c) \ge f(x)$, for all $x \in I$ then f is maximum in I. The point c is known as a point of maximum value in I.



  - **(ii) Minimum value :** If there exists a point c in I such that $f(c) \le f(x)$, for all $x \in I$, then f is minimum in I. The point c is called as a point of minimum value in I.



  - **(iii) Extreme value :** If there exists a point c in I such that $f(c)$ is either a maximum value or a minimum value in I, then f is having an extreme value in I. The point c is said to be an extreme point.

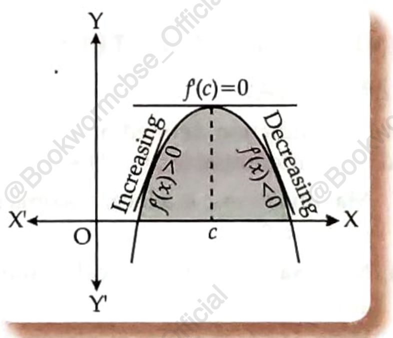


LOCAL MAXIMA AND MINIMA :

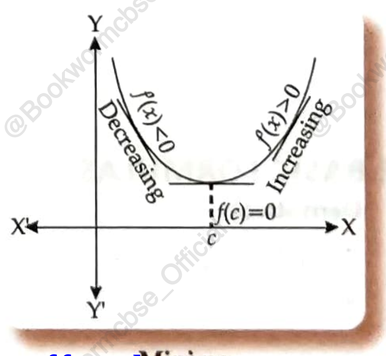
- Let f be a real valued function and c be an interior point in the domain of f, then
  - **(a) Local maxima :** c is a point of local maxima if there is an $h > 0$, $\exists f(c) \ge f(x)\ \forall\ x \in (c - h, c + h)$.
    The value $f(c)$ is called local maximum value of f.
  - **(b) Local minima :** c is a point of local minima if there is an $h > 0$, such that $f(c) \le f(x)$ for all $x \in (c - h, c + h)$.
    The value of $f(c)$ is known as the local minimum value of f.

GEOMETRICALLY :

- If $x = c$ is a point of local maxima of f, then f is increasing ($f'(x) > 0$) in the interval $(c - h, c)$ and decreasing ($f'(x) < 0$) in the interval $(c, c + h)$. This implies $f'(c) = 0$.



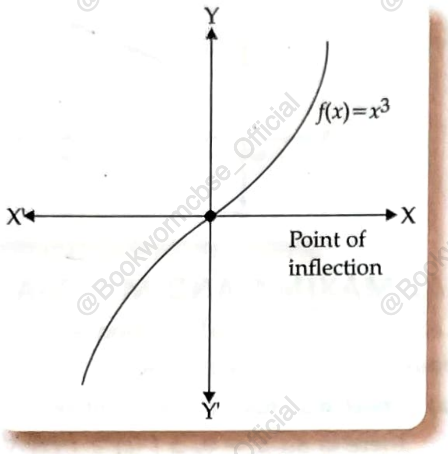
Maxima



Minima

TEST OF MAXIMA AND MINIMA:

- Let f be a function on an open interval I and $c \in I$ be any point. If f has a local maxima or a local minima at $x = c$, then either $f'(c) = 0$ or f is not differentiable at c.



Let f be a continuous at a critical point c in it.

- If $f'(x)$ changes sign from positive to negative as x increases through c i.e.,
  - (i) $f'(x) > 0$ at every point in $(c - h, c)$
  - (ii) $f'(x) < 0$ at every point in $(c, c + h)$

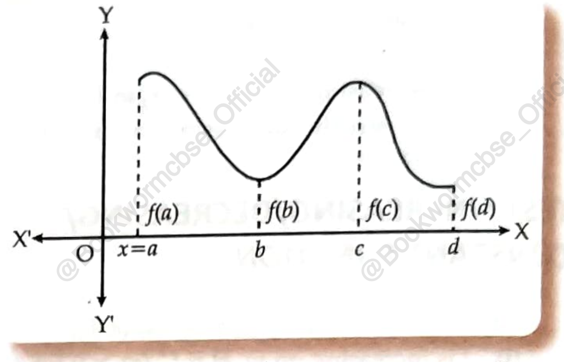
  where h is sufficiently small, there is a point of local maxima.
- If $f'(x)$ changes sign from negative to positive as x increases through c i.e.,
  - (i) $f'(x) < 0$ at every point in $(c - h, c)$
  - (ii) $f'(x) > 0$ at every point in $(c, c + h)$

  where h is sufficiently small. Then, c is a point of local minima.
- If $f'(x)$ does not change sign as x increases through c, then c is neither a point of local maxima nor a point of local minima. Such a point is called point of inflection.

SECOND DERIVATIVE TEST OF MAXIMA AND MINIMA:

- Let f be a function defined on an interval I and $c \in I$ and f be differentiable at c. Then,
  - (i) **Maxima :** $x = c$ is a local maxima if $f'(c) = 0$ and $f''(c) < 0$.
  - (ii) **Minima :** $x = 0$ is a local minima if $f'(c) = 0$ and $f''(c) > 0$. Here, $f(c)$ is the local minimum value of f.
  - (iii) **Point of inflection :** If $f'(c) = 0$ and $f''(c) = 0$, test fails, then we apply first derivative test as x increases through c.

MAXIMUM AND MINIMUM VALUES IN A CLOSED INTERVAL:



Consider the function $f(x) = x + 3, x \in [0, 1]$. Here, $f'(c) \neq 0$. It has neither maxima or minima. But $f(0) = 3$. This is the absolute minimum or global minimum or least value.

Further $f(1) = 4$. This is the absolute maximum or global maximum or greater value.

Further, consider f be any other continuous function having local maxima and minima.

ABSOLUTE MAXIMA AND MINIMA :

Let f be a continuous function on an interval $I = [a, b]$. Then, f has the absolute maximum value and attains at least once in I. Similarly, f has the absolute minimum value and attains at least once in I.

At $x = b$, there is a minima.

At $x = c$, there is a maxima.

At $x = a$, $f(a)$ is the greatest value or absolute maximum value.

At $x = b$, $f(b)$ is the least value or absolute minimum value.

- To find absolute maximum value or absolute minimum value:
  - (i) Find all the critical points *viz.* where $f'(x) = 0$ or f is not differentiable.
  - (ii) Consider the end points also.
  - (iii) Calculate the functional values at all the points found in steps (i) and (ii).
  - (iv) Identify the maximum and minimum values out of the values calculated in step (iii). These are absolute maximum and absolute minimum values.

7. INTEGRALS

SOME BASIC FORMULAS

Derivatives	Integrals (Anti derivatives)
(i) $\frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = x^n$;	$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$

(ii) $\frac{d}{dx}(\sin x) = \cos x$; $\int \cos x dx = \sin x + C$

(iii) $\frac{d}{dx}(\cos x) = -\sin x$; $\int \sin x dx = -\cos x + C$

(iv) $\frac{d}{dx}(\tan x) = \sec^2 x$; $\int \sec^2 x dx = \tan x + C$

(v) $\frac{d}{dx}(\cot x) = -\text{cosec}^2 x$; $\int \text{cosec}^2 x dx = -\cot x + C$

(vi) $\frac{d}{dx}(\sec x) = \sec x \tan x$; $\int \sec x \tan x dx = \sec x + C$

(vii) $\frac{d}{dx}(\text{cosec } x) = -\text{cosec } x \cot x$; $\int \text{cosec } x \cot x dx = -\text{cosec } x + C$

(viii) $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$; $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$

(ix) $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$;

(x) $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$; $\int \frac{dx}{1+x^2} = \tan^{-1} x + C$

(xi) $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$;

(xii) $\frac{d}{dx}(\text{cosec}^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$;

(xiii) $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$; $\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} x + C$

(xiv) $\frac{d}{dx}(e^x) = e^x$; $\int e^x dx = e^x + C$

(xv) $\frac{d}{dx}(\log |x|) = \frac{1}{x}$; $\int \frac{1}{x} dx = \log |x| + C$

(xvi) $\frac{d}{dx}\left(\frac{a^x}{\log a}\right) = a^x$; $\int a^x dx = \frac{a^x}{\log a} + C$

SOME SPECIAL FORMULAS

(i) $\int \tan x dx = \log |\sec x| + C = -\log |\cos x| + C$

(ii) $\int \cot x dx = \log |\sin x| + C$

(iii) $\int \sec x dx = \log |\sec x + \tan x| + C$

(iv) $\int \text{cosec } x dx = \log |\text{cosec } x - \cot x| + C = \log \left|\tan \frac{x}{2}\right| + C$

(v) $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + C$ and $\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + C$

(vi) $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$

(vii) $\int \frac{dx}{\sqrt{x^2-a^2}} = \log\left|x+\sqrt{x^2-a^2}\right|+C$ and $\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left|x+\sqrt{x^2+a^2}\right|+C$

(viii) $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a}+C$

(ix) $\int e^x [f(x) + f'(x)]\, dx = e^x f(x) + C$

(x) $\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right|+C$

and $\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\left|x+\sqrt{x^2+a^2}\right|+C$

(xi) $\int \sqrt{a^2-x^2}\,dx = \frac{1}{2}x\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}+C$

SOME IMPORTANT RULES WHILE DOING INTEGRATION

- In case of rational function, if the degree of numerator is equal or greater than the degree of the denominator, then we first divide the numerator by the denominator and write it as,

$$\frac{\text{Numerator}}{\text{Denominator}} = \text{Quotient} + \frac{\text{Remainder}}{\text{Denominator}} \text{ and then integrate.}$$

- To evaluate $\int \frac{ax+b}{px^2+qx+r}dx$, we write $ax + b = A\frac{d}{dx}(px^2 + qx + r) + B$.
- To evaluate $\int \frac{ax+b}{\sqrt{px^2+qx+r}}dx$, we write $ax + b = A\frac{d}{dx}(px^2 + qx + r) + B$.

INTEGRATION BY PARTIAL FRACTION

Choose the type of partial fraction from the table, find A, B and C, then proceed further by integrating each term separately.

S. No.	Form of the rational function	Form of the partial fraction
1.	$\frac{px+q}{(x-a)(x-b)}, a \neq b$	$\frac{A}{x-a} + \frac{B}{x-b}$
2.	$\frac{px+q}{(x-a)^2}$	$\frac{A}{x-a} + \frac{B}{(x-a)^2}$
3.	$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)}$	$\frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$
4.	$\frac{px^2+qx+r}{(x-a)^2(x-b)}$	$\frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$
5.	$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)}$	$\frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$
	where, $x^2 + bx + c$ cannot be factorised further.	

INTEGRATION BY PARTS

$$\int f(x)\, g(x) dx = f(x) \int g(x) dx - \int [f'(x) \int g(x) dx]\, dx.$$

FUNDAMENTAL THEOREM OF DEFINITE INTEGRAL

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a).$$

SOME PROPERTIES OF DEFINITE INTEGRALS

(i) $P_0 : \int_a^b f(x)dx = \int_a^b f(t)dt$

(ii) $P_1 : \int_b^a f(x)dx = -\int_a^b f(x)dx$, in particular, $\int_a^a f(x)dx = 0$

(iii) $P_2 : \int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$

(iv) $P_3 : \int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

(v) $P_4 : \int_0^a f(x)dx = \int_0^a f(a-x)dx$

(vi) $P_5 : \int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$

(vii) $P_6 : \int_0^{2a} f(x)dx = \begin{cases} 2\int_0^a f(x)dx, \text{ if } f(2a-x) = f(x) \\ 0, \text{ if } f(2a-x) = -f(x) \end{cases}$

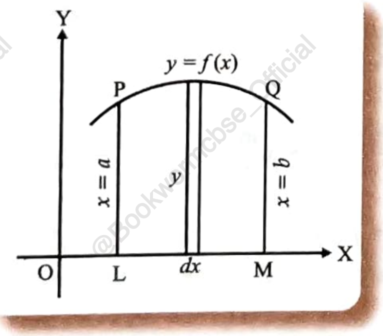
(viii) $P_7 : \int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx$, if f is an even function i.e., $f(-x) = f(x)$

(ix) $P_8 : \int_{-a}^{a} f(x) = 0$, if f is an odd function i.e., $f(-x) = -f(x)$

8. APPLICATIONS OF THE INTEGRALS

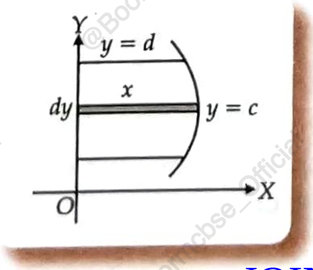
(i) Area under simple curve $y = f(x)$ is given by

$\int_a^b y\, dx = \int_a^b f(x)dx$



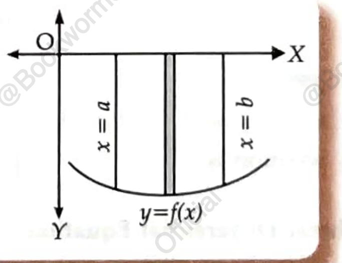
(ii) The area A of the region bounded by the curve $x = g(y)$, y-axis and the lines $y = c$ and $y = d$ is given by

$A = \int_c^d x\, dy = \int_c^d g(y)dy$



(iii) If the curve under consideration lies below X-axis, then $f(x) < 0$ from $x = a$ to $x = b$, the area bounded by the curve $y = f(x)$ and the ordinates $x = a$, $x = b$ and X-axis is negative. But if the numerical value of the area is to be taken into consideration, then.

$$\text{Area} = \left|\int_a^b f(x)dy\right|$$



(iv) It may also happen that some portion of the curve is above X-axis and some portion is below X-axis as shown in the figure. Let A_1 be the area below x-axis and A_2 be the area above the X-axis. Therefore, area bounded by the curve $y = f(x)$, X-axis and the ordinates $x = a$ and $x = b$ is given by

$$A = |A_1| + |A_2|$$

9. DIFFERENTIAL EQUATIONS

FUNDAMENTALS

- An equation involving derivative (derivatives) of the dependent variable with respect to independent variable (variables) is called a differential equation, e.g.,

(i) $x\frac{dx}{dy} + y = 0,$ (ii) $2\frac{d^2y}{dx^2} + y^3 = 0$

- The order of the highest order derivative of dependent variable with respect to the independent variable involved in the differential equation is called the order of the differential equation, e.g.,

$$\frac{dx}{dy} + y = c, \qquad \frac{d^2y}{dx^2} + \frac{dx}{dy} + y = k$$

involve derivatives whose highest orders are 1 and 2 respectively.

- When a differential equation is a polynomial equation in derivatives, the highest power (positive integral index), of the highest order derivative is known as the degree of the differential equation, e.g.,

(i) $\left(\frac{dy}{dx}\right)^2 + \frac{dx}{dy} + y = c$, the highest order derivative is $\frac{dx}{dy}$, its positive integral power is 2.

$\therefore$ Its degree is two.

(ii) $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} + \frac{dx}{dy} + 4 = 0$. Here, the highest order derivative is $\frac{d^3y}{dx^3}$. Its positive integral power is 1.

$\therefore$ Its degree is one.

METHODS OF SOLVING FIRST ORDER, FIRST DEGREE DIFFERENTIAL EQUATIONS

(i) **Variables Separable Method :** "When the equation may be expressed as $\frac{dy}{dx} = h(y)\,g(x)$, then we can write it as $\frac{dy}{h(y)} = g\,(x)dx$.

Integrating, we get the solution as $\int \frac{dy}{h(y)} = \int g\,(x)dx + C$.

(ii) **Homogeneous Differential Equation :** Let us write the differential equation as $\frac{dy}{dx} = f\,(x, y)$.

Replacing x by λx and y by λy and we get $f\,(\lambda x, \lambda y)$

$$= \lambda^n f\,(x, y)$$

Then, the differential equation is homogeneous of degree n.
To solve such an equation

$$\frac{dy}{dx} = f\,(x, y) \text{ put } y = vx \text{ or } \frac{dy}{dx} = v + x\frac{dv}{dx}$$

we get, $$v + x\frac{dv}{dx} = f\,(v) \text{ or } x\frac{dv}{dx} = f\,(v) - v$$

$\therefore$ Solution is $$\int\frac{dv}{f(v)-v} = \int\frac{dx}{x} + C$$

(iii) **Linear Differential Equation :**

(a) The linear differential equation is of the form,

$\frac{dy}{dx} + Py = Q$, where P and Q are the functions of x.

This is a first order, first degree differential equation.
To solve the equation, we find the integrating factor

$$\text{I.F.} = e^{\int P dx}$$

Then, the solution is

$$ye^{\int P dx} = \int Q.e^{\int P.dx}\,dx + C.$$

(b) When the equation is of the form $\frac{dx}{dy} + Px = Q$, where P and Q are the functions of y.

$$\text{I.F.} = e^{\int P dy}$$

$\therefore$ Solution is

$$x e^{\int P dy} = \int Q . e^{\int P dy} \, dy + C.$$

10. VECTORS

FUNDAMENTALS

- A quantity that has magnitude as well as direction is called a vector.
- The unit vector in the direction of $\vec{a}$ is given by $\frac{\vec{a}}{|\vec{a}|}$ and is represented by $\hat{a}$.
- Position vector of a point $P(x, y, z)$ is given as $\overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}$ and its magnitude as $|\overrightarrow{OP}| = \sqrt{x^2 + y^2 + z^2}$, where O is the origin.
- The scalar components of a vector are its direction ratios, and represent its projections along the respective axes.
- The magnitude r, direction ratios (a, b, c) and direction cosines (l, m, n) of any vector are related as:

$$l = \frac{a}{r},\ m = \frac{b}{r},\ n = \frac{c}{r} \quad \text{and} \quad l^2 + m^2 + n^2 = 1$$

- The sum of the vectors representing the three sides of a triangle taken in order is 0.

ADDITION OF VECTORS

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are two vectors then $\vec{a} + \vec{b} = (a_1 + b_1)\hat{i} + (a_2 + b_2)\hat{j} + (a_3 + b_3)\hat{k}$.

SCALAR MULTIPLICATION

If $\vec{a}$ is a given vector and λ is a scalar, then $\lambda\vec{a}$ is a vector whose magnitude is $|\lambda\vec{a}| = |\lambda||\vec{a}|$. The direction of $\lambda\vec{a}$ is same as that of $\vec{a}$ if λ is positive and, opposite to that of $\vec{a}$ if λ is negative.

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are two vectors, then $\lambda\vec{a} = (\lambda a_1)\hat{i} + (\lambda a_2)\hat{j} + (\lambda a_3)\hat{k}$ and

$$\lambda\vec{b} = (\lambda b_1)\hat{i} + (\lambda b_2)\hat{j} + (\lambda b_3)\hat{k}.$$

VECTOR JOINING TWO POINTS

If $P_1\ (x_1, y_1, z_1)$ and $P_2\ (x_2, y_2, z_2)$ are any two points, then

$$\overrightarrow{P_1P_2} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

- Magnitude $= |\overrightarrow{P_1P_2}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

SECTION FORMULA

The position vector of a point R dividing the line segment joining the points P and Q whose position vectors are $\vec{a}$ and $\vec{b}$

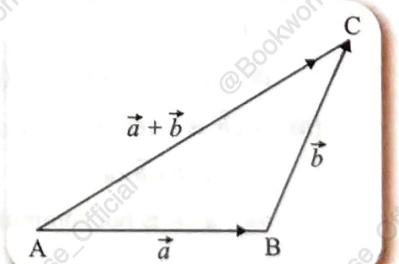
(i) in the ratio $m : n$ internally, is given by $\frac{m\vec{b} + n\vec{a}}{m + n}$

(ii) in the ratio $m : n$ externally, is given by $\frac{m\vec{b} - n\vec{a}}{m - n}$

TRIANGLE LAW OF VECTOR ADDITION

Let the vectors $\vec{a}$ and $\vec{b}$ be so positioned that initial point of one coincides with terminal point of the other, $\vec{a} = \overrightarrow{AB}, \vec{b} = \overrightarrow{BC}$. Then, the vector $\vec{a} + \vec{b}$ is represented by the third side $\overrightarrow{AC}$ of ΔABC i.e.,

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$$



or $$\overrightarrow{AC} - \overrightarrow{AB} = \overrightarrow{BC}$$

or $$\overrightarrow{AC} - \overrightarrow{BC} = \overrightarrow{AB}$$

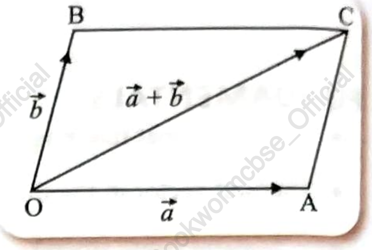
or $$\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \vec{0}$$

This is known as triangle law of addition.

PARALLELOGRAM LAW OF VECTOR ADDITION

If the two vectors $\vec{a}$ and $\vec{b}$ are represented by the two adjacent sides OA and OB of a parallelogram $OACB$, then their sum $\vec{a} + \vec{b}$ is represented by the diagonal $\overline{OC}$ of parallelogram both in magnitude and direction through their common point O.

i.e., $$\overline{OA} + \overline{OB} = \overline{OC}$$



PROJECTION

Projection of vector $\vec{a}$ along vector $\vec{b}$ is $\dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$.

SCALAR OR DOT PRODUCT

The scalar or dot product of two given vectors $\vec{a}$ and $\vec{b}$ having an angle θ between them is defined as:

$$\vec{a}\cdot\vec{b} = |\vec{a}|\,|\vec{b}|\cos\theta$$

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are two vectors, then $\vec{a}\cdot\vec{b} = a_1b_1 + a_2b_2 + a_3b_3$.

PROPERTIES OF SCALAR PRODUCT:

(i) $\vec{a}\cdot\vec{b}$ is a scalar quantity.

(ii) When $\theta = 0$, $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|$

(iii) When $\theta = \dfrac{\pi}{2}$, $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\dfrac{\pi}{2} = 0$

$\Rightarrow$ When $\vec{a} \perp \vec{b}$, $\vec{a}\cdot\vec{b} = 0$

(iv) When either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, $\vec{a}\cdot\vec{b} = 0$

VECTOR OR CROSS PRODUCT

The cross product of two vectors $\vec{a}$ and $\vec{b}$ having angle θ between them is given as

$$\vec{a} \times \vec{b} = |\vec{a}|\,|\vec{b}|\sin\theta\,\hat{n},$$

where $\hat{n}$ is a unit vector perpendicular to the plane containing $\vec{a}$ and $\vec{b}$ If $\vec{a} = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$ and $\vec{b} = a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$ are two vectors, then

$$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = (b_1c_2 - b_2c_1)\hat{i} + (a_2c_1 - a_1c_2)\hat{j} + (a_1b_2 - a_2b_1)\hat{k}$$

PROPERTIES OF VECTOR PRODUCT:

(i) $\vec{a}\times\vec{b} = |\vec{a}|\,|\vec{b}|\sin\theta.\hat{n}$

(a) If $\vec{a} = 0$ or $\vec{b} = 0$, $\vec{a}\times\vec{b} = \vec{0}$

(b) $\vec{a} \,||\, \vec{b}$, $\vec{a}\times\vec{b} = \vec{0}$ (or $\theta = 0$)

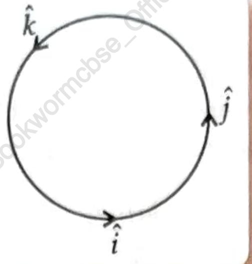
(ii) $\vec{a}\times\vec{b}$ is a vector i.e., $\vec{a}\times\vec{b} = -\vec{b}\times\vec{a}$

$\therefore$ $\vec{a}\times\vec{b} \neq \vec{b}\times\vec{a}$

$\Rightarrow$ $\vec{a}\times\vec{b}$ is not commutative.

(iii) When $\theta = \frac{\pi}{2}$, $\vec{a}\times\vec{b} = |\vec{a}|\times|\vec{b}|\times\hat{n}$

or $|\vec{a}\times\vec{b}| = |\vec{a}||\vec{b}|$



(iv) $\hat{i}\times\hat{i} = \hat{j}\times\hat{j} = \hat{k}\times\hat{k} = \vec{0}$

and $\hat{i}\times\hat{j} = \hat{k}$, $\hat{j}\times\hat{k} = \hat{i}$, $\hat{k}\times\hat{i} = \hat{j}$

$\Rightarrow \hat{j}\times\hat{i} = -\hat{k}, \hat{k}\times\hat{j} = -\hat{i}, \hat{i}\times\hat{k} = -\hat{j}$

(v) $\sin\theta = \dfrac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|}$

(vi) If $\vec{a}$ and $\vec{b}$ represent the adjacent sides of a parallelogram, then its area = $|\vec{a}\times\vec{b}|$.

(vii) If $\vec{a}$ and $\vec{b}$ represents the adjacent sides of a triangle, then its area = $\frac{1}{2}|\vec{a}\times\vec{b}|$.

11. THREE DIMENSIONAL GEOMETRY

FUNDAMENTALS

- Direction cosines of a line are the cosines of the angles made by the line with positive directions of the co-ordinate axes.
- If l, m, n are the direction cosines of a line, then $l^2 + m^2 + n^2 = 1$.
- Direction cosines of a line joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are
- $\dfrac{x_2-x_1}{PQ}, \dfrac{y_2-y_1}{PQ}, \dfrac{z_2-z_1}{PQ}$ where, $PQ = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$
- If l, m, n are the direction cosines and a, b, c are the direction ratios of a line, then

$$l = \frac{a}{\sqrt{a^2+b^2+c^2}},\quad m = \frac{b}{\sqrt{a^2+b^2+c^2}},\quad n = \frac{c}{\sqrt{a^2+b^2+c^2}}$$

- Skew lines are the lines in the space which are neither parallel nor intersecting. They lie in the different planes.
- Angle between skew lines is the angle between two intersecting lines drawn from any point (preferably through the origin) parallel to each of the skew lines.
- Angle between two lines

$$\cos\theta = |l_1l_2 + m_1m_2 + n_1n_2|$$

or

$$\cos\theta = \left|\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+a_2^2+a_3^2}.\sqrt{b_1^2+b_2^2+b_3^2}}\right|$$

LINE

- Vector equation of a line that passes through the given point whose position vector is $\vec{a}$ and parallel to a given vector $\vec{b}$ is $\vec{r} = \vec{a}+\lambda\vec{b}$
- Equation of a line through a point (x_1, y_1, z_1) and having direction cosines l, m, n (or, direction ratios a, b and c) is

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} \quad \text{or} \quad \left(\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}\right)$$

- The vector equation of a line that passes through two points whose position vectors are $\vec{a}$ and $\vec{b}$ is $\vec{r} = \vec{a}+\lambda(\vec{b}-\vec{a})$.
- Cartesian equation of a line that passes through two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

- If θ is the acute angle between the lines $\vec{r} = \vec{a}_1+\lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2+\lambda\vec{b}_2$, then θ is given by

$$\cos\theta = \frac{|\vec{b}_1.\vec{b}_2|}{|\vec{b}_1||\vec{b}_2|} \quad \text{or} \quad \theta = \cos^{-1}\frac{|\vec{b}_1.\vec{b}_2|}{|\vec{b}_1||\vec{b}_2|}.$$

- If $\frac{x-x_1}{l_1}=\frac{y-y_1}{m_1}=\frac{z-z_1}{n_1}$ and $\frac{x-x_2}{l_2}=\frac{y-y_2}{m_2}=\frac{z-z_2}{n_2}$ are equations of two lines, then the acute angle θ between the two lines is given by $\cos\theta = |l_1l_2 + m_1m_2 + n_1n_2|$.
- The shortest distance between two skew lines is the length of the line segment perpendicular to both the lines.
- The shortest distance between the lines $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\lambda\vec{b}_2$ is

$$\left|\frac{(\vec{b}_1\times\vec{b}_2)\cdot(\vec{a}_2-\vec{a}_1)}{|\vec{b}_1\times\vec{b}_2|}\right|$$

- Shortest distance between the lines : $\frac{x-x_1}{a_1}=\frac{y-y_1}{b_1}=\frac{z-z_1}{c_1}$ and $\frac{x-x_2}{a_2}=\frac{y-y_2}{b_2}=\frac{z-z_2}{c_2}$ is

$$d=\frac{\begin{vmatrix}x_2-x_1 & y_2-y_1 & z_2-z_1\\ a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\end{vmatrix}}{\sqrt{(b_1c_2-b_2c_1)^2+(c_1a_2-c_2a_1)^2+(a_1b_2-a_2b_1)^2}}$$

- Distance between parallel lines $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\lambda\vec{b}_2$ is $d=\left|\frac{\vec{b}\times(\vec{a}_2-\vec{a}_1)}{|\vec{b}|}\right|$ (if $\vec{b_1}=\lambda\vec{b_2}$)

12. LINEAR PROGRAMMING

- A linear programming problem deals with the optimization (maximization/ minimization) of a linear function of two variables (say x and y) known as objective function subject to the conditions that the variables are non-negative and satisfy a set of linear inequalities (called linear constraints). A linear programming problem is a special type of optimization problem.
- **Objective Function :** Linear function, $Z = ax + by$, where a and b are constants, which has to be maximised or minimized is called a linear objective function.
- **Decision Variables :** In the objective function, $Z = ax + by$, x and y are called decision variables.
- **Constraints :** The linear inequalities or restrictions on the variables of an LPP are called constraints. The conditions $x \ge 0$, $y \ge 0$ are called non-negative constraints.
- **Feasible Region :** The common region determined by all the constraints including non-negative constraints $x \ge 0$, $y \ge 0$ of an LPP is called the feasible region for the problem.
- **Feasible Solutions :** Points within and on the boundary of the feasible region for an LPP represent feasible solutions.
- **Infeasible Solutions :** Any point outside feasible region is called an infeasible solution.
- **Corner point method for solving a LPP :**
  The method comprises of the following steps :
  (i) Find the feasible region of the LPP and determine its corner points (vertices) either by inspection or by solving the two equations of the lines intersecting at that point.
  (ii) Evaluate the objective function $Z = ax + by$ at each corner point.
  Let M and m, respectively denote the largest and the smallest values of Z.
  (iii) (a) When the feasible region is bounded, M and m are respectively, the maximum and minimum values of Z.
  (b) In case, the feasible region is unbounded.
    - M is the maximum value of Z, if the open half plane determined by $ax + by > M$ has no point in common with the feasible region. Otherwise, Z has no maximum value.
    - Similarly, m is the minimum of Z, if the open half plane determined by $ax + by < m$ has no point in common with the feasible region. Otherwise, Z has no minimum value.

13. PROBABILITY

CONDITIONAL PROBABILITY

If E and F are two events associated with the same sample space of a random experiment, then the conditional probability of the event E under the condition that the event F has occurred, written as $P(E \mid F)$, is given by

$$P(E \mid F) = \frac{P(E\cap F)}{P(F)},\ P(F) \ne 0$$

PROPERTIES OF CONDITIONAL PROBABILITY

Let E and F be events associated with the sample space S of an experiment. Then :

(i) $P(S|F) = P(F|F) = 1$

(ii) $P[(A \cup B) \mid F] = P(A \mid F) + P(B \mid F) - P[(A \cap B \mid F)]$,
where A and B are any two events associated with S.

(iii) $P(E' \mid F) = 1 - P(E \mid F)$

MULTIPLICATION THEOREM ON PROBABILITY

Let E and F be two events associated with a sample space of an experiment. Then

$$\begin{aligned} P(E \cap F) &= P(E)\, P(F \mid E),\ P(E) \neq 0 \\ &= P(F)\, P(E \mid F),\ P(F) \neq 0 \end{aligned}$$

If E, F and G are three events associated with a sample space, then

$$P(E \cap F \cap G) = P(E)\, P(F \mid E)\, P(G \mid E \cap F)$$

INDEPENDENT EVENTS

Let E and F be two events associated with a sample space S. If the probability of occurrence of one of them is not affected by the occurrence of the other, then we say that the two events are independent. Thus, two events E and F will be independent, if

(i) $P(F \mid E) = P(F)$, provided $P(E) \neq 0$

(ii) $P(E \mid F) = P(E)$, provided $P(F) \neq 0$

Using the multiplication theorem on probability, we have

(iii) $P(E \cap F) = P(E)\, P(F)$

Three events A, B and C are said to be mutually independent if all the following conditions hold :

$$\begin{aligned} P(A \cap B) &= P(A)\, P(B) \\ P(A \cap C) &= P(A)\, P(C) \\ P(B \cap C) &= P(B)\, P(C) \\ \text{and} \quad P(A \cap B \cap C) &= P(A)\, P(B)\, P(C) \end{aligned}$$

PARTITION OF A SAMPLE SPACE

A set of events $E_1, E_2, \ldots, E_n$ is said to represent a partition of a sample space S if

(i) $E_i \cap E_j = \phi,\ i \neq 0;\ i, j = 1, 2, 3, \ldots\ldots, n$

(ii) $E_1 \cup E_2 \cup \ldots \cup E_n = S$ and

(iii) Each $E_i \neq \phi$, i.e., $P(E_i) > 0$ for all $i = 1, 2, \ldots, n$

THEOREM OF TOTAL PROBABILITY

Let $\{E_1, E_2, \ldots, E_n\}$ be a partition of the sample space S. Let A be any event associated with S, then

$$P(A) = \sum_{j=1}^{n} P(E_j) P(A|E_j)$$

BAYES' THEOREM

If $E_1, E_2, \ldots, E_n$ are mutually exclusive and exhaustive events associated with a sample space, and A is any event of non-zero probability, then

$$P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum_{i=1}^{n} P(E_i)P(A|E_i)}$$